

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		6.45 cm ³		1		1		
		(ii)		moles thiosulfate = $0.500 \times \frac{6.45}{1000} = 3.225 \times 10^{-3}$ mol (1) reaction ratio $\Rightarrow 2\text{S}_2\text{O}_3^{2-} \equiv \text{I}_2 \equiv 1 \text{ClO}^-$ so moles chlorate(I) = 1.6125×10^{-3} mol (1) concentration chlorate(I) = $\frac{1.6125 \times 10^{-3}}{25 \times 10^{-3}} \times 10 = 0.645$ mol dm ⁻³ (1) ecf possible from part (i)		3		3	2	
		(iii)		$0.645 \times \frac{74.5}{10} = 4.81\%$		1		1	1	
		(iv)		should have chosen 0.200 mol dm ⁻³ marks credited for reasons, MAX 1 mark for reasons if a different concentration is chosen use a lower concentration so titration volume is greater – smaller percentage error / more accurate (1) cannot use too low a concentration / 0.0500 mol dm ⁻³ as volume would be too large (for a standard burette) (1)			2	2		2
	(b)	(i)	I	order with respect to $\text{ClO}_3^- \Rightarrow$ first order (1) order with respect to $\text{Br}^- \Rightarrow$ first order (1) must show working to gain each mark		2		2	2	

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	(c)	(i)		K_a of stronger acid would be <u>greater</u> than that for weaker acid as stronger acids have <u>greater dissociation</u>	1			1		
		(ii)	I	Joe's method: rearrangement of $pV = nRT$ (1) $n = 3.5 \times 10^{-3}$ (1) accept alternative method Heledd's method: $n = 3.48 \times 10^{-3}$ or 3.484×10^{-3} (1) must show use of M_r of carbon dioxide to gain mark both answers to appropriate significant figures and both concentrations are the same (ECF from calculations) (1)		1 1		4	3	1
			II	Heledd's method as measurements are to more significant figures / more precise / have higher resolution			1	1		1
			III	would make Heledd's experiment less accurate as the mass released would be much smaller (as H_2 has much smaller M_r than CO_2) (1) will not affect the accuracy of Joe's experiment as same volume of gas would be produced (1)			1 1	2		
		(iii)		pH from 2-6 (1) must attempt reason to gain this mark award (1) for either of following ammonium ion (partially) dissociates to release H^+ ions $NH_4^+ \rightleftharpoons NH_3 + H^+$ do not accept compound/salt/ammonium perchlorate releases H^+	1	1		2		
Question 11 total					2	14	8	24	13	4